Aidan Hart Ray Pine

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Education

University of Edinburgh — Edinburgh, UK
M.Sc. Speech & Language Processing — *September 2020 – September 2021*

- Dissertation title: "Low Resource Speech Synthesis"
- Distinction (80% coursework average, 90% Dissertation grade)
- Saltire Scholarship (£8,000)

University of British Columbia — Vancouver, BC
B.A. Honours Linguistics, Minor in First Nations' Languages — *September 2012 – April 2016*

- 4.21 GPA (90.2% Average)
- Sole recipient, Dean's Leadership Award for Outstanding Academic Contributions to the Faculty of Arts
- 1st Place Oral Presentation at UBC Multidisciplinary Undergraduate Research Conference 2016

Experience

National Research Council Canada — *Victoria, BC*

- Associate Research Scientist — July 2024 – present
- Assistant Research Scientist — July 2023 – July 2024
- Indigenous Languages Technologist — January 2022 – July 2023
- Application Development Specialist — January 2018 – January 2022

Software Developer & DevOps Engineer — October 2016 – June 2020
First Peoples' Heritage and Language Council of Canada — *Victoria, BC*

- Developed 18 digital dictionary apps for iOS and Android, custom approximate search algorithms and continuous deployment devops architecture with fastlane

Software Developer — November 2016 – June 2019
University of British Columbia — *Vancouver, BC*

- Lead programmer for the development of four digital 'talking' dictionaries for web, iOS, and Android, including language-customized approximate search algorithms

Software Developer — August 2016 – January 2017
Simon Fraser University — *Vancouver, BC*

- Lead developer for development of orthography conversion tools for Haíłzaqv and Tsilhqot'in

Community, Leadership & Service

Member, ISO/IEC JTC 1/SC 2 (Coded Character Sets) — April 2024 – present
Standards Council of Canada — *Remote*

- Canadian mirror committee member providing input to ISO/IEC 10646, the international standard counterpart to Unicode
- Advocates for the character encoding needs of Indigenous languages, a prerequisite for any written language to be used digitally

Volunteer Developer — May 2016 – present
Various — *Remote Work*

- Volunteer supporting a variety of online open-source projects including:
- Squamish Dictionary project, Peter Jacobs, Simon Fraser University
- Inupiaq Dictionary project, Myles Creed, University of Victoria
- Designed and built Chrome OS keyboard for SENĆOŦEN language

Reviewer — Periodically since 2018
Various — *Remote Work*

- Transactions on Audio, Speech and Language Processing (TASLP)
- Computer Speech & Language (CS&L)

- Journal of Medical Internet Research (JMIR)
- Expert Systems With Applications (ESWA)
- Native American and Indigenous Studies Journal (NAIS)
- Association for Computational Linguistics (ACL)
- ELRA Language Resources and Evaluation Conference (LREC)
- The Use of Computational Methods in the Study of Endangered Languages Workshop (ComputEL)

Selected Publications

Tse, T., Zhu, J., **Pine, A.**, Geng, M. 2026
AudioNoisePrints: Model-free audio watermarking using spatial correlation in flow matching TTS.
Proceedings of *Interspeech 2026*.

Pine, A., Lothian, D., Bird, S., Caldecott, M., MENEŦIYE, PENÁĆ, Richmond, K., Swallow, T., SX̱EDŦELISIYE, et al. 2025
Practical and Contextual Speech Synthesis Evaluation.
Proceedings of *Speech Synthesis Workshop 2025*.

Pine, A., Cooper, E., Guzmán, D., Joanis, E., Krekoski, R., Kuhn, R., Larkin, S., Littell, P., Lothian, D., et al. 2025
Speech Generation for Indigenous Language Education.
Computer Speech and Language.

Littell, P., Joanis, E., **Pine, A.**, Tessier, M., Huggins-Daines, D., Torkornoo, D. 2022
ReadAlong Studio: Practical Zero-Shot Text-Speech Alignment for Indigenous Language Audiobooks.
Proceedings of *SIGUL 2022*. Marseille, France.

Pine, A., Littell, P., Joanis, E., Huggins-Daines, D., Cox, C., Davis, F., Santos, E., Srikanth, S., Torkornoo, D., et al. 2022
G_i2P_i: Rule-based, index-preserving grapheme-to-phoneme transformations.
Proceedings of *ComputEL-5*. Dublin, Ireland.

Pine, A., Wells, D., Brinklow, N., Littell, P., Richmond, K. 2022
Requirements and motivations of low-resource speech synthesis for language revitalization.
Proceedings of *ACL 2022*. Dublin, Ireland.
***Awarded 'Best Theme Paper'**

Brinklow, N., Littell, P., Lothian, D., **Pine, A.**, Souter, H. 2020
Indigenous Language Technologies and Language Reclamation in Canada.
Proceedings of *International Conference Language Technologies for All (LT4All): Enabling Linguistic Diversity and Multilingualism Worldwide*.

Kazantseva, A., Maracle, B., Maracle, R., **Pine, A.** 2018
Kawennón:nis: the Wordmaker for Kanyen'kéha (Ohsweken Mohawk).
Proceedings of *COLING: The 27th International Conference on Computational Linguistics*. Santa Fe, New Mexico, USA.

Pine, A., Turin, M. 2017
Language Revitalization.
Oxford Research Encyclopedia of Linguistics. Oxford University Press.

Teaching

Guest Lectures 2015–2024

- Over a dozen guest lectures taught at 1st - 4th year undergraduate classes at the University of British Columbia (FNEL180, 382 & ASTU402), University of Victoria (LING372), and University of Waterloo (Mohawk 101R)

"Watch me Speak!" Interactive Storytelling using Read-Along Studio 2022

- **Pine, A.**, Huggins-Daines, D., Joanis, E., Littell, P., Tessier, M., Torkornoo, D. ICLDC 8 Workshop. University of Hawaii.

How to Make a 'Mother Tongues' Digital Dictionary 2019

- **Pine, A.**, Turin, M. ICLDC 6 Workshop. University of Hawaii.

Digital Lexicography 2018

- **Pine, A.**, Turin, M. Course taught at CoLang Summer Institute 2018 at the University of Florida.

A course in Practical Phonetics 2017, 2018

- Invited to teach a series of practical phonetics courses to two Gitksan community language classes in Gitsegukla, BC and Gitanmaax, BC

Developing Digital Tools for Language Revitalization: Demystifying Coding, Apps, and Web Platforms 2018

- **Pine, A.**, Turin, M. Invited multi-day workshop at the National Institute for Japanese Language and Linguistics. Tokyo.

Python for Lexicographers Workshop Series 2016

- Taught basic Python and programming skills to undergraduate, graduate and post-graduate lexicographers at the UBC linguistics department

Doreen Jensen Memorial Gitksan Class 2013–2016

- Teaching assistant. Explained grammatical, phonological and orthographic concepts to students

Awards & Grants

Digital Government Community Award, Excellence in Inclusion, Diversity, Equity & Accessibility (Indigenous Language Technology team award, Treasury Board of Canada Secretariat) 2026

ESRC Impact Acceleration Grant (£29,809 + £38,682 in-kind) 2026
Own Your Voice: Community-Led Solutions for Underrepresented Languages

Ideation Fund Small Teams Initiative ($1,875,150) 2022–2025
Speech Generation for Indigenous Language Education

National Research Council Digital Technology 'Value for Canada' Award 2024

NRC IP Achievement Award (as part of Indigenous Language Technology Project) ($6,000) 2022

ACL2022 Best Special Theme Paper ($1,000) 2022

National Research Council Digital Technology 'Research & Technology Breakthrough' Award 2022

Scotland's Saltire Scholarship (£8,000) 2020

National Research Council Digital Technology 'Value for Canada' Award 2018

Sanderson Family Service Award 2016

1st Place Oral Presentation at UBC Multidisciplinary Undergraduate Research Conference 2016

Dean's Leadership Award for Outstanding Academic Contributions to the Faculty of Arts 2016

Jacobs Research Fund 2014, 2015 & 2016

UBC Arts Undergraduate Research Award 2015

UBC Student Scholarship in Arts 2015

UBC Arts Undergraduate Research Award 2014

UBC Work Learn 2013 & 2014

Technical Skills

Machine Learning: Speech synthesis, data efficient (low-resource) modelling, grapheme-to-phoneme conversion, text-audio alignment, speech recognition, PyTorch, PyTorch Lightning
Speech Evaluation: listening-test design, qualitative and quantitative methods, statistical analysis
Programming: Python, JavaScript, TypeScript
Web Development: FastAPI, Angular, StencilJS, HTML/CSS/Sass
Signal Processing: Praat, NumPy, librosa, ffmpeg
Infrastructure: CI/CD, Slurm, Hugging Face, Docker
Mobile Development: iOS and Android development with Ionic, fastlane

Languages

Danish: Intermediate/Advanced. Prøve i Dansk 3 Fluency Certificate.
French: Intermediate/Advanced. Double Dogwood High School Diploma in English & French.
Gitksan: Beginner/Intermediate.
hən̓q̓əmin̓əm: Beginner/Intermediate. UBC FNLG 101, 102, 201, 202, & 301.
SENĆOŦEN: Beginner. UVic LING 159.
nêhiyawêwin: Beginner. UBC FLNG 111.